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Q.1- Explain about Arithmetic and Logic unit.

Ans 1- Arithmetic and logic unit (ALU) is a vital component of CPU responsible for performing various arithmetical and logical operations. Typically, ALU has direct input and output access to the processor controller, main memory (RAM) and input/output (I/O) devices. Inputs and outputs flow along an electronic path is called a bus.

The input consists of an instruction word, sometimes called a machine instruction word, that contains an operation code or "opcode", one or more operands and sometimes a format code. The opcode informs ALU about which operation to be performed and the operands used in the operation.

ALU serve as a combinational digital circuit that performs arithmetic and bitwise operations on binary numbers.

Some operations supported by ALU —

Addition, subtraction, Increment, decrement, AND, OR, NOT, XOR, etc.

Q.2- Define ALU, control unit, input and output unit.

Ans 2- ALU : Arithmetic Logic unit (ALU) is the tool of CPU because of which a computer can perform arithmetic operations (like +, -, *, /, etc.) and Logical operations (like AND, OR, NOT, XOR, NAND, etc.)

Control unit : Control unit or CU is that component of CPU which directs all the CPU-operations. Its job is to instruct ALU, memory parts, input & output devices on how to respond to the instructions supplied on the processor.

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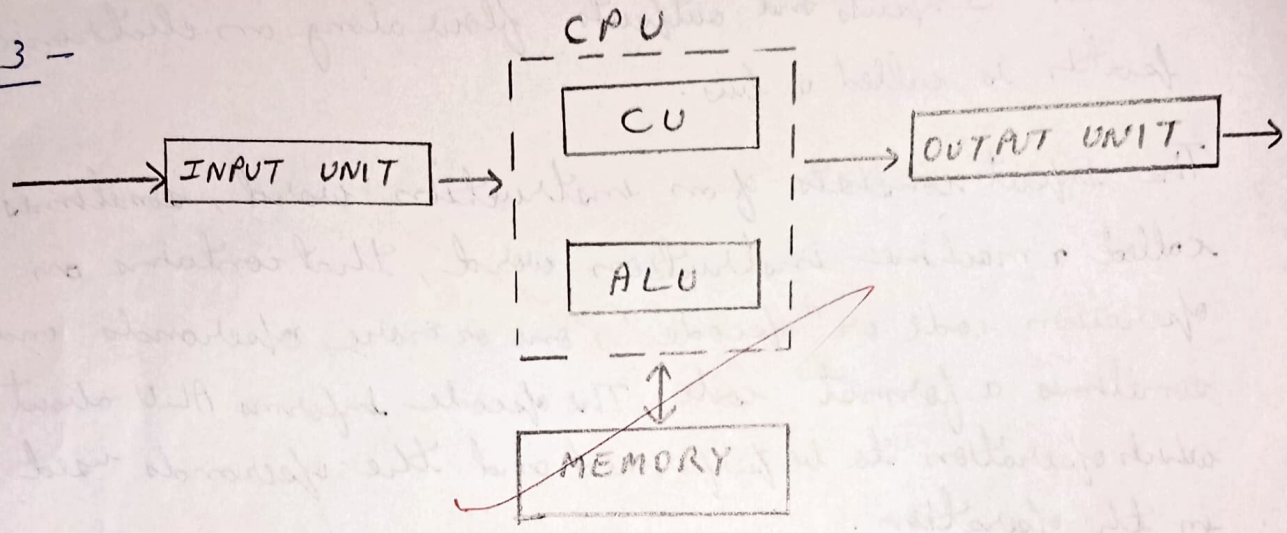
Input and output unit (I/O unit):

Input unit is the unit from which user can send/feed/write data to a computer.
output unit is the unit from where some processed data is received by user, sent from the computer.

Q.3 - Discuss basic functional units of computer in details.

Ans 3 -



There are five main (or basic) functional units of computer —

(i) Input unit : This unit takes data from the user and allows the system to understand the task on that data by further sending it to the CPU.

(ii) output unit : The only purpose of output unit is to convey/display the result of processed input entered by the user.

(iii) ALU : Arithmetic-Logic unit or ALU is responsible for solving all kind of arithmetical & Logical problems encountered by the computer. Along with this, ALU can also store the results of operations in the CPU's registers (temporarily).

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iv) CU : Control unit or CU is an essential part of CPU which controls the data flow, sent or received by the CPU. It acts as a coordinator, responsible for accepting inputs, storing data in the memory, processing it if requires, displaying the data in the output unit.

(v) Memory : The main objective of memory unit is to store data and instructions. There are mainly 3 types of memory — Primary (RAM, ROM), Secondary and Cache memory.

Q.4- Explain about below registers :
• MAR • MDR • PC • GPR • IR

Ans - (i) MAR — Memory Address Register (MAR) contains the address of a location of main memory from where information has to fetched / stored. Contents of MAR are directly connected to the address bus.

(ii) MDR — Memory Data register (MDR) contains the data which is either to be written into the memory or recently being read from the memory. It is directly connected to the data bus. Also called Memory Buffer register (MBR).

(iii) PC — Program counter (PC) contains the address of an instruction to be executed next. After every execution the content of PC is updated so that it always points to the next instruction to be executed.

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(iv) GPR — General Purpose registers (GPRs) can store both data and addresses. These are extra registers that are present in the CPU and are utilized anytime when data or memory location is required. Mainly used to store operands for arithmetic & Logic operations.

(v) IR — Instruction Register (IR) contains the instruction which is most recently fetched or executed. The fetched instruction is loaded into an IR, where the opcode and operand specifier is analyzed.

Q.5 - Explain single bus and multi-bus structure.

Ans 5 - Single bus System

As goes by the name Single bus system has only one bus that connects all the components of the system. Here, all the data and instructions are transferred between the components through a single bus, though it can limit the performance of the system. Commonly used in small or simpler computer systems.

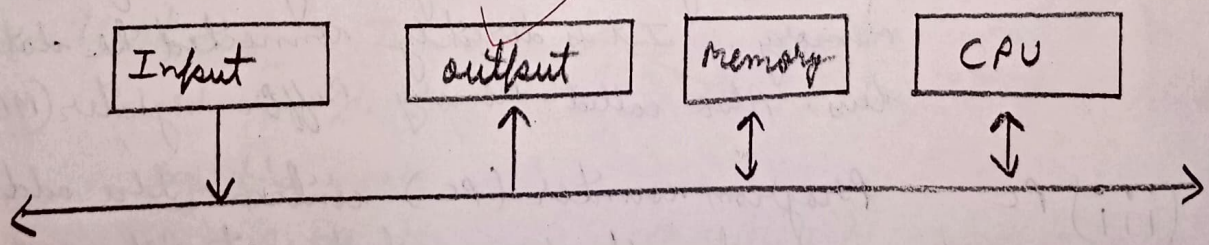

Fig : Single Bus System

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Multi-Bus System

A Multi-Bus System has multiple buses that connect different components of the system. Each bus can have a specific function, such as data transfer, memory access or I/O operations. This allows the system for more efficient communication (among components) and reduces the risk of entire system failure. Multi-Bus systems are employed in larger, complex computer systems.

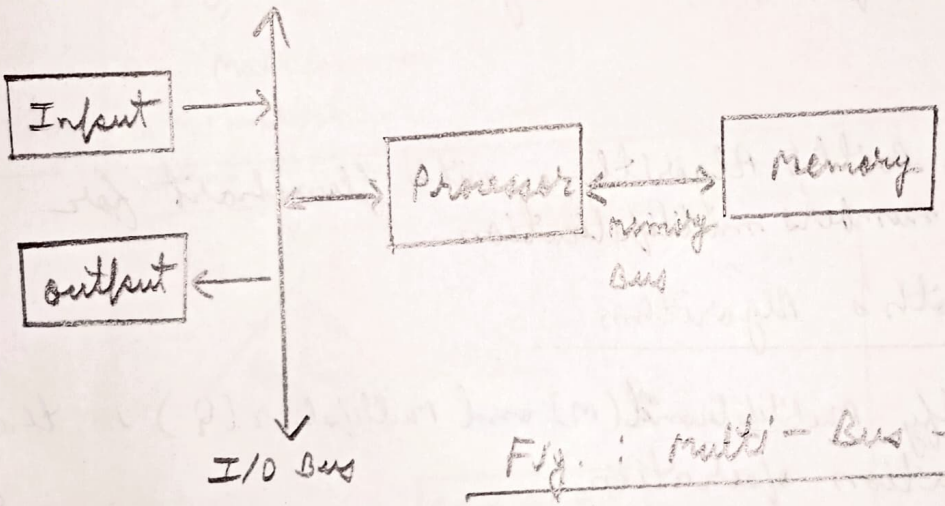

Fig. : Multi-Bus System

Q.6- Explain address bus, data bus and control bus.

Ans 6 - (i) Address bus

- The address bus carries information about the memory location or I/O port where data needs to be read from or written to.
- Its width determines the maximum addressable memory space. A wider address bus can access a larger memory.

(ii) Data bus

- The Data bus is responsible for transmitting data between the CPU, memory and input/output devices.
- It is bi-directional (allowing data to flow in both directions)

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- The width of the data bus determines the amount of data that can be transmitted simultaneously.

E.g: 32-bit data bus ⇒ 32 bits of data at a time.

(iii) Control Bus

- The control bus carries control signals that coordinate and regulate the activities of various components.
- Control signals include read and write signals, interrupt requests, clock signals and bus control signals.

Q.7 - Write Booth's Algorithm with flowchart for signed numbers multiplication.

Ans 7 - Booth's Algorithm

1.) Identify Multiplicand (M) and Multiplier (Q) in the Multiplication operation.

2.) Declare and initialize 3 variables AC (accumulator) as 0, Q_{-1} as 0 and count as n (no. of bits in AC). Also, Declare Q_0 such that Q_0 = LSB of Q.

3.) As per the corresponding values of Q_0 & Q_{-1}, perform the below mentioned operation:

Q_0	Q_{-1}	
0	0	→ ① ASR
0	1	→ ① AC = AC + M, ② ASR
1	0	→ ① AC = AC − M = AC + (−M), ② ASR
1	1	→ ① ASR

4.) After performing each ASR (i.e., Arithmetic shift right) operation, decrement count by 1 i.e.,

count = count − 1.

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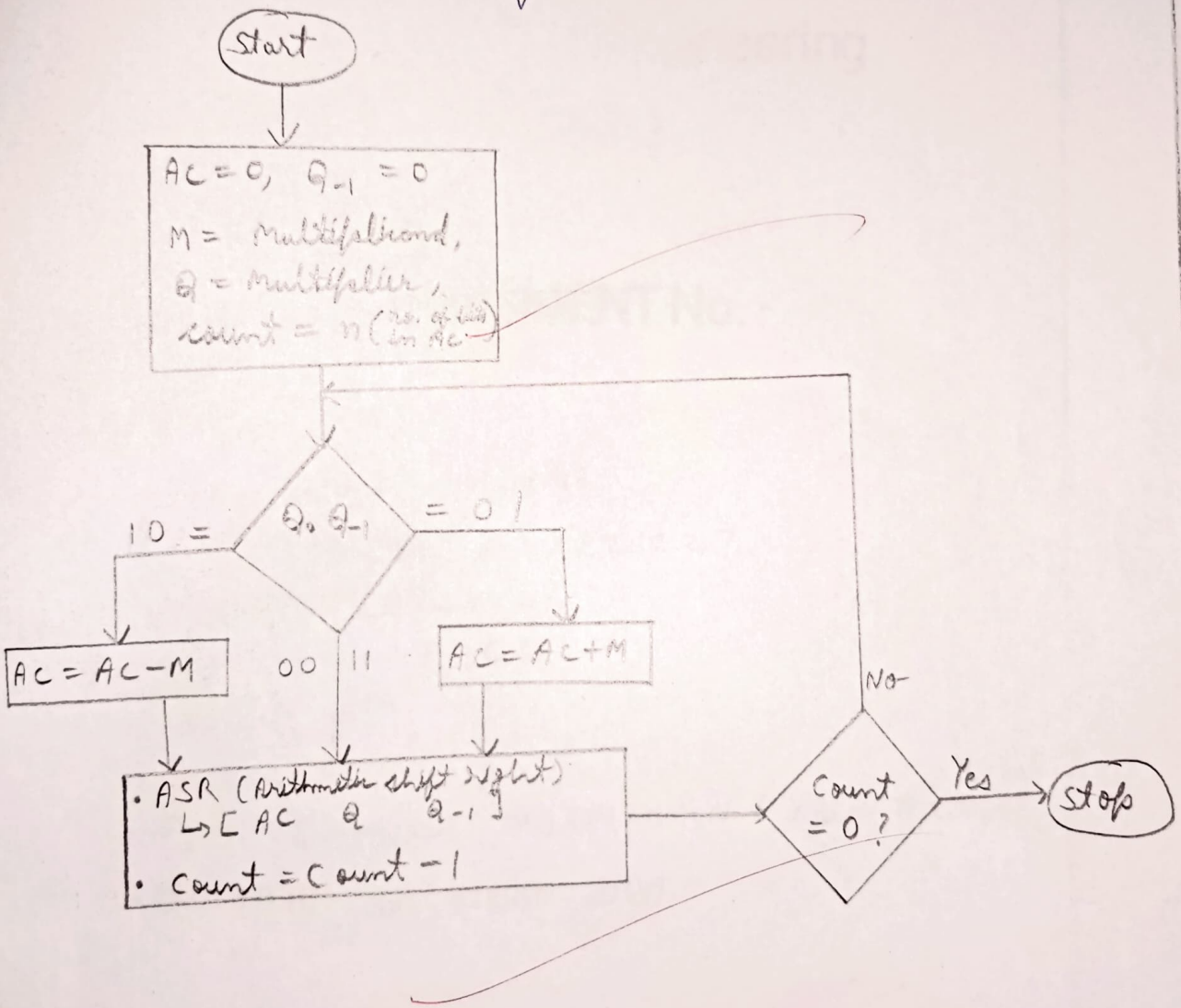
Repeat steps 3 to 4, until count becomes 0. When count becomes 0, halt the procedure.

6. Now, the value of AC and Q (simply joined together) gives the answer to the multiplication.

Flowchart for Booth's Algorithm



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